PROBLEM SET 4

1.) Calculate $\ln\Lambda$ for a plasma with $n_e = n_i = 10^{20}\,m^{-3}$ and $T_e = 5\,keV$ for:

a.) A high energy electron beam of energy $E = 100\,keV$ colliding with stationary Hydrogen beams.

Velocity of Incident electrons: $v_e = \sqrt{\frac{2K}{m}}$

This gives us: $\Lambda = \left(\frac{\epsilon_0 T_e}{ne^2}\right)^{1/2}\left(\frac{q_1 q_2 m_e}{4\pi\epsilon_0 m_r 2K_e}\right)^{1/2}$, $\frac{m_e}{m_r} = \frac{m_e + m_i}{m_i} \approx 1$

I am too lazy to calculate the actual Numerical Values.

b.) This beam colliding with the electrons in this plasma.

We evaluate this with the thermal relative velocity: $\sqrt{\frac{2K}{m_e}} - \sqrt{\frac{T_e}{m_e}}$

c.) The thermal collisions in this plasma:

Here for $i \to e$ collisions: $v = \sqrt{\frac{T_e}{m_e}}$

$\downarrow$

Stationary.

For $e \to e$ or $i \to e$ collisions: $v = \sqrt{\frac{T_e}{m_r}}$